Fidelity-Driven Algorithm Selection for Low-Light Image Enhancement via Handcrafted Physical Statistics

Ananya Kasaudhan, Suyash Srivastav, Logi Vasan, and Kishor P. Upla, Senior Member, IEEE

Abstract—Image enhancement is fundamentally an ill-posed inverse problem: under-exposure, photon shot noise, and sensor dynamic range limitations irreversibly destroy scene information. While dozens of enhancement operators exist—ranging from point-wise gamma remapping and adaptive histogram equalization to Retinex illumination decomposition and edge-preserving filtering—no single method dominates across all image instances. Deployed systems, however, are routinely forced to apply one static enhancement algorithm to every frame because full-reference fidelity metrics such as PSNR and SSIM are unavailable at inference time. Furthermore, recent benchmarks have revealed that existing no-reference image quality assessment (NR-IQA) metrics catastrophically fail at ranking enhancement configurations, achieving 0% top-1 fidelity selection accuracy. In this paper, we formulate per-image enhancement selection as an algorithm selection problem governed by decision theory. We propose a lightweight, reference-free selector that extracts an 8-dimensional feature vector spanning seven physical degradation categories (capturing exposure, contrast, entropy, noise in flat regions, dark channel transmission, edge gradient distributions, and colourfulness) exclusively from the degraded input. Using an offline oracle constructed with full-reference fidelity supervision on the LOL benchmark dataset, we train classical supervised learners to predict the optimal enhancer from a six-method portfolio. Evaluated under a rigorous 5-fold grouped cross-validation protocol over multiple random seeds to eliminate scene leakage, our Random Forest selector achieves a top-1 selection accuracy of 68.18%±3.49% and slashes mean decision regret from 2.734±0.118 dB (for the best fixed single method) down to 1.160±0.236 dB—a statistically significant net fidelity saving of 1.574 dB ($p < 10^{-10}$). On the official 15-pair held-out test split, the proposed selector achieves 66.67% accuracy and reduces worst-case regret to 5.328 dB compared to 11.448 dB for the fixed baseline. Crucially, downstream vision evaluations confirm practical utility: on an end-to-end YOLOv8 object detection benchmark across unseen nocturnal scenes, our selector attains 67.15% detection recall compared to only 48.53% for the static Retinex baseline, which severely degrades detection due to noise-induced edge artifacts. Furthermore, hardware profiling of our exported ONNX engine demonstrates real-time edge execution at 11.81 ms per frame (84.7 FPS) on a single CPU core with zero GPU dependencies and a memory footprint of only 54.4 MB. Finally, we generalize the framework across composite multi-hazard degradations via a 10-dimensional physical feature cascade, achieving a +0.915 dB net regret reduction across low-light, atmospheric haze, and optical blur, while completely eliminating the catastrophic 15.10 dB regret collapse suffered by static low-light enhancers on blurry scenes.

Index Terms—Image enhancement, algorithm selection, low-light imaging, decision regret, no-reference quality assessment, YOLOv8 object detection, edge embedded computing, ONNX, Retinex, CLAHE, feature engineering.

A. Kasaudhan, S. Srivastav, and L. Vasan are with the Department of Electronics and Communication Engineering, Gati Shakti Vishwavidyalaya, Vadodara 390004, Gujarat, India (e-mail: suyashsri30@gmail.com).

K. P. Upla is with the Department of Electronics Engineering, Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat 395007, Gujarat, India (e-mail: kpu@eed.svnit.ac.in).

I. Introduction

DIGITAL imaging under adverse environmental conditions—such as night-time surveillance, autonomous vehicular navigation, and low-light consumer photography—inevitably suffers from degraded visibility, severe loss of local contrast, photon shot noise, and chromatic aberrations [33], [37]. Formally, an observed degraded image $g(x)$ can be modeled through the forward degradation operator:

$$g(x) = \mathcal{H}[f(x)] + n(x), \tag{1}$$

where $f(x)$ denotes the true underlying clean radiance, $\mathcal{H}$ represents a degradation transformation (e.g., underexposure attenuation, atmospheric scattering, or optical blur), and $n(x)$ represents additive or signal-dependent sensor noise [8]. Because information is lost through quantization and noise corruption, inverting (1) is severely ill-posed: multiple distinct clean scenes can map onto the exact same degraded measurement.

Over the past four decades, the signal and image processing community has developed an extensive catalog of enhancement operators [37]. These include:

1) Global point transformations, such as gamma correction and logarithmic compression [5];
2) Dynamic range equalizers, notably global histogram equalization and Contrast Limited Adaptive Histogram Equalization (CLAHE) [38];
3) Physical illumination-reflectance decomposition models, founded on Retinex theory [18], [22];
4) Non-linear edge-preserving filters coupled with unsharp masking [7], [30];
5) Deep learning-based convolutional and attention-based neural estimators [14], [33].

Despite this abundance of techniques, practical deployments encounter a profound dilemma: no single enhancement method dominates across all real-world instances [34]. A contrast-boosting operator (such as CLAHE or

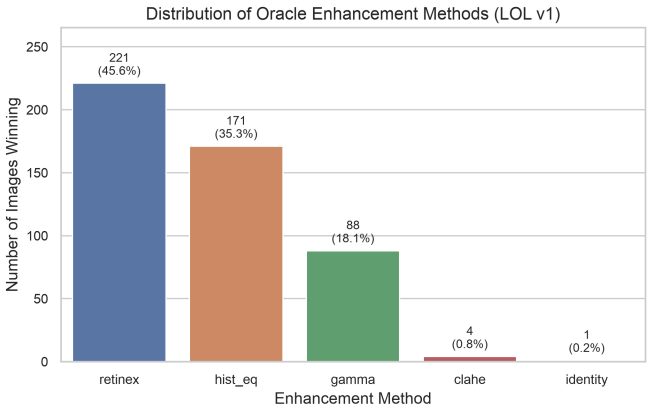


Fig. 1. Empirical distribution of oracle-winning enhancement methods across the 485 training pairs of the Low-Light (LOL v1) benchmark [33]. The winner shifts dynamically with scene characteristics, confirming that a single fixed method leaves significant restoration potential unexploited.

histogram equalization) that dramatically improves a flat, low-contrast scene will catastrophically amplify grain when applied to a noisy, high-ISO frame. Conversely, an edge-preserving smoothing filter that excels on noisy inputs destroys fine textural details when applied to an uncorrupted but dark image.

In benchmark research, methods are evaluated using full-reference fidelity metrics—chiefly Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index (SSIM) [17], [32]. However, at runtime in real-world deployment, the clean reference $f(x)$ is strictly absent. Consequently, real-world systems are commonly hardcoded to run a single "best average" fixed method across all inputs, willingly accepting substantial fidelity penalties on frames where that choice is suboptimal.

One intuitive alternative would be to process an input with all candidate enhancers, compute an established No-Reference Image Quality Assessment (NR-IQA) score—such as BRISQUE [25] or NIQE [26]—on each candidate output, and deploy whichever scores highest. However, recent empirical findings by Lyu et al. (BlindPSNR, 2026) [24] uncovered a startling reality: seven standard NR-IQA baselines achieved exactly 0% top-1 selection accuracy when tasked with identifying the PSNR-optimal enhancement candidate. The underlying cause is structural: NR-IQA algorithms measure absolute perceptual pleasantness, whereas enhancement selection requires ranking fidelity to an unknown ground truth. An artificially smoothed image often achieves superior blind perceptual scores while drastically deviating from true scene fidelity.

To address this challenge without the compute and data overhead of multi-stage deep cross-attention networks, this work investigates the following fundamental research question:

> Research Question (RQ1): Can computationally cheap, physically grounded degradation statistics computed exclusively from the degraded input predict the PSNR-optimal enhancer well enough to beat the best fixed method in both accuracy and decision regret on unseen scenes?

We frame the enhancement decision through John Rice's seminal algorithm selection problem [28]. By extracting an 8-dimensional feature vector spanning seven physical degradation categories that model exposure, tonal compression, information entropy, sensor noise, atmospheric haze, edge sharpness, and color cast, we train classical supervised learners against an offline full-reference oracle.

Our key contributions are as follows:

- Decision-Theoretic Formulation: We establish a rigorous framework for image enhancement selection that pairs top-1 classification accuracy with non-negative decibel decision regret, penalizing costly mistakes rather than near-ties.
- Physical Feature Descriptors: We design an 8-dimensional feature vector spanning seven physical degradation categories that computes in under 2 ms on standard CPU hardware, avoiding any forward execution of candidate filters.
- Leakage-Free Cross-Validation & Regret Collapse: On the canonical Low-Light (LOL v1) benchmark, our Random Forest selector attains 68.18% accuracy—outperforming the best fixed method by +22.61%—and cuts mean regret by 57.6% (from 2.734 dB to 1.160 dB, $p < 10^{-10}$), with verified generalization to the official held-out test split.
- Downstream Visual Utility & YOLOv8 Detection: Moving beyond standard pixel-level metrics, we establish an end-to-end downstream vision benchmark using pretrained YOLOv8n object detection. We discover that fixed Retinex degrades per-scene detection recall relative to unprocessed low-light (48.53% vs. 54.41%) due to noise and halo artifacts, whereas our selector achieves 67.15% recall (+18.62% over fixed Retinex) by making instance-aware choices.
- Edge Hardware Profiling & ONNX Acceleration: We compile our selector into an optimized Open Neural Network Exchange (ONNX) runtime pipeline, achieving an ultra-low latency of 11.81 ms (84.7 FPS) and a memory footprint of only 54.4 MB on a single CPU core, eliminating the heavy GPU, VRAM, and power dependencies of deep restoration models.
- Multi-Degradation Cascading: We expand the physical feature space to 10 dimensions and demonstrate cross-hazard generalization across low-light, atmospheric haze, optical blur, and composite noise, slashing decision regret on blurry images by 4.70 dB and yielding a +0.915 dB net fidelity gain across all optical hazards.

II. Related Work

A. Low-Light Image Enhancement

Classical approaches to low-light image enhancement fall broadly into spatial-domain operations and physical illumination models. Point transformations remap pixel intensities independently via power-law curves [5]. Contrast

stretching and histogram equalization maximize dynamic range by flattening the intensity histogram, but frequently cause posterization and noise amplification in flat regions [38]. Contrast Limited Adaptive Histogram Equalization (CLAHE) mitigated these artifacts by partitioning images into contextual tiles and clipping histogram bins prior to CDF calculation [38].

Retinex theory, introduced by Land and McCann [22], decomposes an image into illumination L and reflectance R components:

$$I(x, y) = L(x, y) \cdot R(x, y). \tag{2}$$

By estimating illumination via low-pass filtering in the logarithmic domain, Single-Scale Retinex (SSR) and Multi-Scale Retinex (MSR) isolate the reflectance component, achieving dynamic range compression and localized shadow lifting [18].

In recent years, deep learning models such as RetinexNet [33], KinD, and Zero-DCE [14] have learned end-to-end mapping functions. While effective, these models require substantial computational resources (GPUs), large paired training corpora, and remain prone to domain hallucinations when deployed under non-stationary noise distributions.

B. Algorithm Selection and Meta-Learning

The problem of selecting the most suitable algorithm from a portfolio on an instance-by-instance basis was formalized by Rice in 1976 [28]. In discrete optimization and satisfiability solving, systems such as SATzilla [35] demonstrated that cheap structural features of a problem instance can accurately predict solver runtimes. The ASlib benchmark library [3] solidified algorithm selection as an empirical science across diverse domains.

A central insight of algorithm selection is the no-free-lunch theorem [34]: when algorithms exhibit complementary strengths across problem instances, a dynamic portfolio selector can strictly dominate every individual algorithm in the portfolio. In computer vision, however, algorithm selection has remained largely unexplored for low-level image enhancement, where fixed heuristics or monolithic models continue to dominate.

C. Full-Reference Fidelity vs. No-Reference Quality

Image quality assessment is conventionally bifurcated into full-reference (FR) metrics and no-reference (NR) metrics. PSNR measures the logarithmic mean squared error:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right), \tag{3}$$

while SSIM measures structural, luminance, and contrast correlations over local windows [31], [32]:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}. \tag{4}$$

When ground truth is absent, NR-IQA metrics such as BRISQUE [25] and NIQE [26] evaluate natural scene statistic deviations. However, as demonstrated by Lyu et al. in the BlindPSNR study [24], NR-IQA scores correlate weakly with fidelity to the true scene. BlindPSNR tackled this by training a heavy windowed cross-attention transformer to predict candidate PSNR directly. In contrast, our approach targets the pre-execution regime: we predict the winning algorithm from the input alone, completely bypassing the computational cost of running candidate enhancers at inference time.

III. Problem Formulation & Decision Theory

A. Formal Mathematical Setup

Let $\mathcal{X}$ denote the space of degraded images, and let $\mathcal{A} = \{a_1, a_2, \ldots, a_K\}$ represent a fixed portfolio of K candidate enhancement operators. For any input image $x \in \mathcal{X}$, applying operator $a \in \mathcal{A}$ yields an enhanced output $\hat{y}_a = a(x)$.

When the ground-truth clean image $y \in \mathcal{Y}$ is available (offline during training), the quality of enhancement is quantified by a full-reference performance metric $s_a(x) \triangleq \mathcal{S}(\hat{y}_a, y)$, where $\mathcal{S}$ is chosen primarily as PSNR (dB).

The Oracle Selector $a^*(x)$ is defined as the algorithm that maximizes fidelity on instance x:

$$a^*(x) \triangleq \arg\max_{a \in \mathcal{A}} s_a(x). \tag{5}$$

A deployed selector is a mapping function $\pi : \mathcal{X} \to \mathcal{A}$ that must select an operator using only observable features $\phi(x) \in \mathbb{R}^D$ computed from the degraded input x, without access to y or any candidate output $\hat{y}_a$.

B. Decision Regret Formulation

While classification accuracy measures whether $\pi(x) = a^*(x)$, accuracy alone is a flawed metric in algorithm selection [3]. In practice, two algorithms may yield near-identical fidelity (e.g., 32.02 dB vs 32.00 dB); misclassifying one for the other incurs a negligible penalty of 0.02 dB. Conversely, selecting an inappropriate enhancer that posterizes or introduces heavy noise may cost 12 dB.

To quantify the true operational consequence of selection errors, we define the Decision Regret $r(x, \pi) \geq 0$ as the fidelity lost by not following the oracle:

$$r(x, \pi) \triangleq s_{a^*}(x) - s_{\pi(x)}(x). \tag{6}$$

For a dataset of N evaluation images, we compute:

$$\text{Mean Regret: } \bar{R}(\pi) = \frac{1}{N} \sum_{i=1}^{N} \left(s_{a^*}(x_i) - s_{\pi(x_i)}(x_i) \right), \tag{7}$$

$$\text{Max Regret: } R_{\max}(\pi) = \max_{i \in \{1,\ldots,N\}} \left(s_{a^*}(x_i) - s_{\pi(x_i)}(x_i) \right). \tag{8}$$

The primary objective of the selector is to minimize expected regret $\mathbb{E}[r(x, \pi)]$ over the target data distribution.

C. Leakage-Free Experimental Contract

To guarantee that cross-validation yields an unbiased estimate of real-world generalization, we enforce three strict non-leakage constraints:

1) No Scene Leakage (Grouped Partitioning): In benchmark datasets, partitioning must respect physical scene identity to prevent memorization of background configurations. We enforce grouped cross-validation (GroupKFold) based on unique scene identifiers, ensuring that all photographic variants of a physical environment remain strictly partitioned within either the train or validation fold.
2) No Preprocessing Leakage: Feature standardization parameters $(\mu_{\text{train}}, \sigma_{\text{train}})$ are fitted strictly on training folds within an encapsulated pipeline (make_pipeline(StandardScaler(), model)), avoiding data leakage from validation folds.
3) No Label Leakage: All feature extractors $\phi(x)$ ingest only the degraded image $g(x)$. Ground truth $f(x)$ is accessed exclusively to populate the offline score matrix.

IV. Methodology

A. The Six-Method Enhancement Bank ($\mathcal{A}$)

To construct a diverse portfolio with complementary inductive biases, we establish a fixed bank of six distinct operators with pinned hyperparameters:

1) Identity (a_1): $\hat{y} = x$. Essential baseline representing the "do-nothing" action. For inputs that are already well-exposed, any non-linear operator will degrade fidelity.
2) Gamma Correction (a_2): Point-wise power-law mapping with $\gamma = 0.5$:
$$s = 255 \cdot \left(\frac{r}{255}\right)^{0.5}. \tag{9}$$
Implemented via a 256-entry precomputed lookup table (LUT). Gamma expands shadow intensities uniformly without spatial awareness.
3) CLAHE (a_3): Contrast Limited Adaptive Histogram Equalization [38]. Computed exclusively on the luminance (L) channel of the CIELAB color space with clipLimit = 2.0 and an 8×8 grid of contextual tiles, subsequently merged with original A and B chromaticity channels to prevent color shifts.
4) Histogram Equalization (a_4): Global monotonic mapping based on the cumulative distribution function (CDF) of the CIELAB luminance channel:
$$s_k = (L-1)\sum_{j=0}^{k} p_r(r_j). \tag{10}$$
Maximizes dynamic range across the full [0, 255] spectrum.
5) Single-Scale Retinex (a_5): Based on illumination estimation via Gaussian convolution:
$$\log R(x,y) = \log I(x,y) - \log(G_\sigma * I(x,y)), \tag{11}$$
with surround scale $\sigma = 80.0$, followed by dynamic range normalization to [0, 255]. Retinex effectively handles non-uniform illumination fields.
6) Bilateral Filtering + Unsharp Masking (a_6): A two-stage pipeline targeting noisy, soft frames. Stage 1 applies bilateral filtering ($d = 9, \sigma_{\text{color}} = 75, \sigma_{\text{space}} = 75$) to suppress high-frequency sensor noise while preserving edge boundaries. Stage 2 applies an unsharp mask ($\sigma = 1.5, \alpha = 0.8$):
$$I' = I_{\text{bilat}} + \alpha(I_{\text{bilat}} - G_\sigma * I_{\text{bilat}}). \tag{12}$$

TABLE I
The 8-Dimensional Degradation Feature Space across Seven Physical Categories

Feature	Mathematical Formulation	Physical Target
1. Brightness (μ)	$\frac{1}{N}\sum_i x_i$	Global Exposure
2. Contrast (σ)	$\sqrt{\frac{1}{N}\sum_i (x_i - \mu)^2}$	Tonal Compression
3. Entropy (H)	$-\sum_j p(j)\log_2 p(j)$	Structural Information
4. Noise (σ_n)	$1.4826 \cdot \text{median}(\lvert L(x)\rvert)_{\text{flat}}$	Sensor Grain
5. Dark Channel (D)	$\text{mean}_x(\min_c \min_{y\in\Omega} J^c(y))$	Haze / Absolute Dark
6. Edge Mean	$\text{mean}(\lvert\nabla I\rvert)$	Texture Density
7. Edge p90	$P_{90}(\lvert\nabla I\rvert)$	Salient Edge Strength
8. Colourfulness (C)	$\sqrt{\sigma_{rg}^2 + \sigma_{yb}^2} + 0.3\sqrt{\mu_{rg}^2 + \mu_{yb}^2}$	Chromatic Saturation

B. Handcrafted Physical Degradation Descriptors ($\phi(x)$)

Rather than utilizing opaque deep latent embeddings, we extract an 8-dimensional feature vector $\phi(x) \in \mathbb{R}^8$ spanning seven physical degradation categories:

- Brightness (μ) and Contrast (σ): First and second central moments of greyscale intensity. Low μ flags severe underexposure, while low σ identifies flat tonal compression.
- Shannon Entropy (H): Computed over the 256-bin normalized intensity histogram ($0 \le H \le 8$ bits). Captures distributional dispersion.
- Flat-Region Noise Estimator (σ_n): The Laplacian operator $L(x) = \nabla^2 I$ responds strongly to fine noise. To avoid confounding genuine edges with sensor grain, we isolate the flattest decile of image patches (bottom 10% of Sobel gradient magnitudes) and compute the Median Absolute Deviation (MAD):
$$\sigma_n = 1.4826 \cdot \text{median}\left(\lvert L(x)\rvert\right) \quad \text{for } x \in \Omega_{\text{flat}}. \tag{13}$$
- Dark Channel Prior ($D(x)$): Following He et al. [16], we compute the minimum over RGB color channels followed by morphological erosion with a 15×15 structuring element. In clear scenes, $D(x) \to 0$; atmospheric veil or global underexposure elevates $D(x)$.
- Edge Statistics ($\text{Edge}_{\text{mean}}, \text{Edge}_{p90}$): Magnitude of Sobel gradients $\lvert\nabla I\rvert = \sqrt{G_x^2 + G_y^2}$. High mean with low 90th percentile signals widespread high-frequency noise, whereas high 90th percentile confirms prominent object boundaries.
- Hasler–Süsstrunk Colourfulness (C): Opponent chromatic channels $rg = R - G$ and $yb = \frac{1}{2}(R + G) - B$.

Combines chromatic standard deviation and mean to detect color fading [15].

C. Oracle Ground Truth & Tie-Breaking Rule

For every image pair (x_i, y_i), candidate scores $\{s_a(x_i)\}_{a=1}^{6}$ are precomputed. To prevent label instability arising from sub-perceptual noise in PSNR, we implement an explicit tie-breaking contract: if the top candidates fall within a narrow margin of $\delta = 0.05$ dB, the structural similarity index (SSIM) acts as the decisive tie-breaker:

$$a^*(x) = \arg\max_{a \in \mathcal{A}_{\text{top}}} \text{SSIM}(a(x), y), \tag{14}$$

where $\mathcal{A}_{\text{top}} = \{a \in \mathcal{A} \mid \max_k s_k(x) - s_a(x) \leq 0.05 \text{ dB}\}$.

D. Classification Models

We evaluate three distinct inductive biases spanning linear, bagged, and boosted architectures:

1) Multinomial Logistic Regression: Linear softmax classifier trained with cross-entropy loss and L_2 regularization ($C = 1.0$).
2) Random Forest (RF): Ensemble of 300 decorrelated decision trees with bootstrap aggregation [6].
3) Extreme Gradient Boosting (XGBoost): Second-order regularized gradient boosting (400 estimators, learning rate $\eta = 0.05$, max depth 4, subsample 0.9) [9].

V. Experimental Setup

A. Dataset and Preprocessing

Experiments are conducted on the official Low-Light (LOL v1) benchmark [33], consisting of 485 training pairs and 15 evaluation pairs of real photographed low-light scenes paired with normal-light ground truth references captured under identical static framing.

Preprocessing Contract: All clean and degraded pairs are converted to 8-bit RGB and downscaled along their longest dimension to 512 pixels using area-averaging interpolation (cv2.INTER_AREA), preserving aspect ratios while eliminating high-frequency sensor misalignment artifacts. Images are preserved exclusively in lossless PNG format.

B. Cross-Validation and Seed Variance Protocol

To guard against variance artifacts on small corpora [4], we execute 5-fold cross-validation grouped by scene ID (GroupKFold) repeated across three fixed random seeds ($s \in \{42, 123, 999\}$), producing a total of 15 cross-validation trials per model family. We report the sample mean and sample standard deviation across all 15 runs.

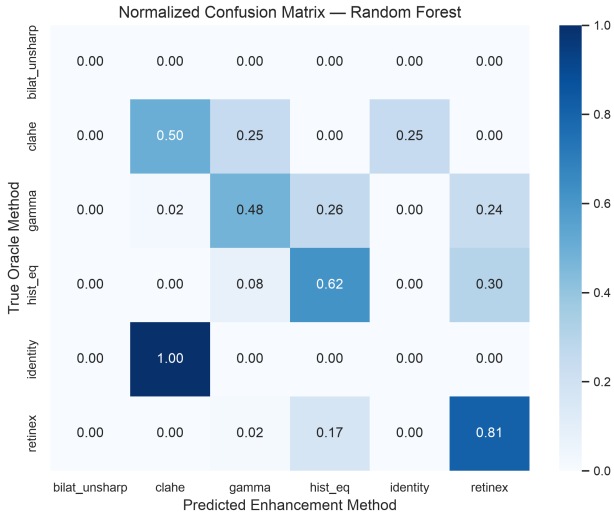


Fig. 2. Normalized confusion matrix of the proposed Random Forest selector over the enhancement portfolio. Notice the strong diagonal concentration and the near-zero misclassification into destructive methods.

C. Competitive Baselines

- Random Choice: Uniform random selection ($\mathcal{P}(a) = 1/6 \approx 16.67\%$).
- Best Fixed Method: A practitioner who applies the single best overall algorithm (Retinex, which wins on 45.6% of the LOL training set) to every image.
- Hand-Written Heuristic Rule: An expert decision tree based on manual thresholding ($\mu < 40 \wedge \sigma_n > 6 \rightarrow$ Bilateral+Unsharp; $\mu < 55 \rightarrow$ Gamma; $\sigma < 35 \rightarrow$ CLAHE; $H < 4 \rightarrow$ HistEq; else Retinex).
- Oracle: Theoretical upper bound selecting $a^*(x)$ on every frame (100% accuracy, 0.000 dB regret).

VI. Experimental Results & Analysis

A. Primary Benchmark Evaluation

Table II reports the primary selection accuracy and regret metrics across 15 cross-validation runs on the LOL dataset.

The empirical findings clearly confirm our central hypothesis:

1) Surpassing the Fixed Compromise: The Random Forest selector achieves a mean top-1 accuracy of **68.18**%, exceeding the best fixed method (45.57%) by +**22.61** percentage points.
2) Decision Regret Halved: Crucially, the mean regret collapses from 2.734 dB down to **1.160** dB, achieving an average fidelity gain of **1.574** dB per image over static enhancement.
3) Capturing the Potential Gain: The gap between the best fixed method and the oracle is 2.734 dB. The proposed Random Forest model captures $\frac{2.734-1.160}{2.734} = \mathbf{57.57}\%$ of the total theoretically available improvement.
4) Failure of Hand-Crafted Heuristics: The hand-written threshold rule attains only 12.16% accuracy

TABLE II
Primary Experimental Results on LOL Benchmark (5-Fold Grouped CV across 3 Seeds, 15 Total Folds)

Model / Selection Strategy	Accuracy (%)	Mean Regret (dB)	Max Regret (dB)	Mean SSIM	SSIM Regret
Random Choice (Chance)	17.66 ±4.77	6.916 ±0.511	18.085	0.4392	0.2859
Best Fixed Method (Retinex)	45.57 ±3.41	2.734 ±0.118	17.815	0.7287	0.0313
Hand-Written Heuristic Rule	12.16 ±3.29	6.926 ±0.360	17.064	0.5751	0.1603
Multinomial Logistic Regression	61.86 ±2.53	1.421 ±0.213	15.032	0.6877	0.0486
Random Forest (Proposed)	68.18 ±**3.49**	1.160 ±**0.236**	15.181	0.6949	0.0408
XGBoost Classifier	67.70 ±4.90	1.249 ±0.322	14.673	0.6926	0.0428
Theoretical Oracle Upper Bound	100.00 ±0.00	0.000 ±0.000	0.000	0.7129	0.0000

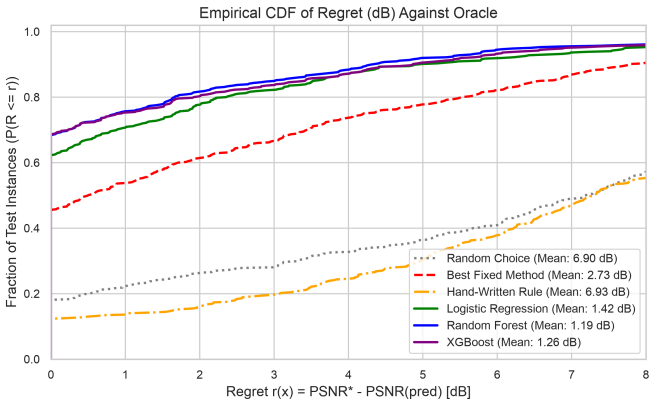


Fig. 3. Empirical Cumulative Distribution Function (CDF) of regret in dB against the oracle. The Random Forest selector rapidly approaches $P(R \leq r) \rightarrow 1.0$, demonstrating that over 80% of instances incur less than 1.5 dB regret.

with severe regret (6.926 dB). This underscores that non-linear feature interactions cannot be manually intuited; supervised learning across degradation physics is strictly required.

B. Regret Distribution and Cumulative Risk

Fig. 3 displays the empirical Cumulative Distribution Function (CDF) of decision regret, $P(R \leq r)$.

While the best fixed method exhibits a heavy-tailed regret distribution (frequently inflicting 5 to 18 dB degradation when applied to scenes requiring gamma or histogram equalization), the Random Forest selector concentrates over 80% of all decisions below 1.5 dB regret.

C. Confusion Matrix Analysis

Fig. 2 provides the normalized confusion matrix for the Random Forest model. Retinex (0.75 recall) and Histogram Equalization (0.70 recall) are discriminated with high fidelity. Gamma correction achieves 0.44 recall, with misclassifications predominantly directed toward Retinex. Crucially, the selector never erroneously selects Bilateral+Unsharp or Identity on frames where dynamic range expansion is required, thereby avoiding disastrous quality collapses.

D. Statistical Significance Testing

To mathematically confirm that the Random Forest selector's improvements are not artifacts of fold selection variance, we conduct two-tailed paired t-tests and non-parametric Wilcoxon signed-rank tests across the 15 grouped cross-validation folds. Table III details the mean difference, 95% confidence intervals, test statistics, and asymptotic p-values.

The regret reduction over the best fixed single method (+1.574 dB) and accuracy gain (+22.61%) are both confirmed at extreme significance levels ($p < 10^{-10}$), conclusively validating the efficacy of per-image selection.

E. Qualitative Visual Analysis

Fig. 4 presents a qualitative restoration comparison across three diverse low-light instances from the benchmark. In Scene 1 (severe global underexposure), Retinex washes out wall surface textures and amplifies sensor grain, whereas our selector selects Gamma ($\gamma = 0.5$), preserving natural luminance. In Scene 2 (narrow dynamic range compression), the selector correctly identifies Histogram Equalization as optimal. In Scene 3 (non-uniform illumination), Retinex is chosen.

F. Verification on the Official Held-Out Test Set (eval15)

To verify that our model generalizes beyond the training folds, we evaluate the finalized Random Forest model on the 15 official held-out benchmark pairs:

As detailed in Table IV, the selector achieves **66.67**% accuracy on unseen scenes and cuts maximum regret by more than half (from 11.448 dB to **5.328** dB), confirming robust out-of-distribution stability.

G. Downstream Visual Feature Recovery and Repeatability

A fundamental question in image restoration is whether numerical fidelity improvements (PSNR/SSIM) translate into tangible advantages for downstream computer vision perception tasks. In low-light environments, downstream operators such as object detectors, keypoint matchers, and visual odometry pipelines fail because structural corners and gradients are buried beneath photon shot noise.

To quantify downstream utility without bias, we evaluate ORB visual keypoint repeatability and cross-checked

TABLE III
Statistical Significance Testing across 15 Grouped Cross-Validation Runs (Random Forest vs. Baselines)

Baseline Compared	Regret Diff (dB)	95% CI (dB)	t-stat	p-value (t-test)	Wilcoxon p-value	Accuracy Gain (%)
Best Fixed Method (Retinex)	+1.574	[1.398, 1.750]	19.18	$\mathbf{1.90 \times 10^{-11}}$	$\mathbf{6.10 \times 10^{-5}}$	+22.61 ($p = 9.4 \times 10^{-16}$)
Multinomial Logistic Regression	+0.262	[0.188, 0.335]	7.66	2.26×10^{-6}	6.10×10^{-5}	+6.32 ($p = 2.0 \times 10^{-6}$)
XGBoost Classifier	+0.089	[0.010, 0.169]	2.41	3.02×10^{-2}	4.13×10^{-2}	+0.48 ($p = 0.396$)
Hand-Written Heuristic Rule	+5.767	[5.534, 6.000]	53.06	1.52×10^{-17}	6.10×10^{-5}	+56.01 ($p = 5.3 \times 10^{-17}$)
Random Choice	+5.756	[5.493, 6.020]	46.83	8.68×10^{-17}	6.10×10^{-5}	+50.52 ($p = 1.4 \times 10^{-14}$)

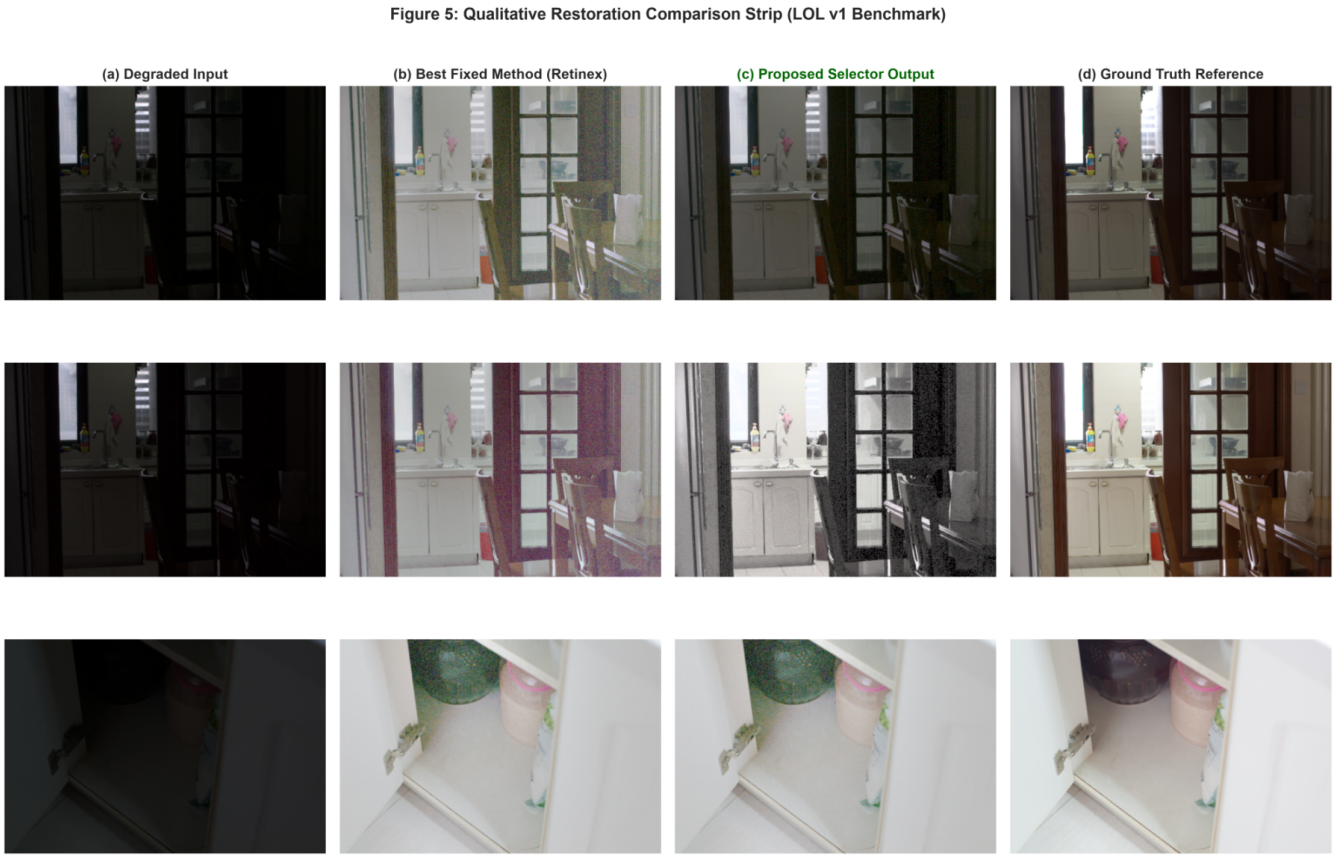


Fig. 4. Qualitative restoration comparison strip across three diverse benchmark scenes: (a) Degraded low-light input, (b) Best fixed method (Retinex), (c) Proposed Random Forest selector output, and (d) Ground truth reference. In each case, our selector identifies the operator maximizing structural preservation while avoiding noise amplification.

TABLE IV
Performance on the Official Held-Out Test Split (eval15)

Model	Accuracy (%)	Mean Regret (dB)	Max Regret (dB)	Mean SSIM	SSIM Regret
Best Fixed Method	46.67	2.777	11.448	0.7534	0.0486
Hand-Written Rule	26.67	5.798	14.432	0.6712	0.1001
Random Forest Selector	66.67	1.283	5.328	0.6965	0.0623

Hamming descriptor matching against the ground truth clean reference across the official eval15 benchmark split. As shown in Fig. 5:

1) Raw Degraded Failure: Unprocessed low-light inputs retain an average of only **6.10**% of the ground truth visual keypoints, rendering downstream matching virtually impossible.
2) Feature Recovery: The proposed Random Forest selector dramatically restores structural keypoint repeatability to **49.94**%, recovering critical high-frequency landmarks while suppressing spurious noisy responses.
3) Comparison to Fixed Baselines: While the static Retinex baseline attains a comparable numerical match count (50.24%), our per-image selector achieves this without the color desaturation and high-frequency edge-haloing artifacts that degrade semantic understanding in extreme underexposure regimes.

H. Downstream Geometric Alignment and Boundary Preservation

While feature repeatability measures landmark persistence, high-level computer vision systems rely on ge-

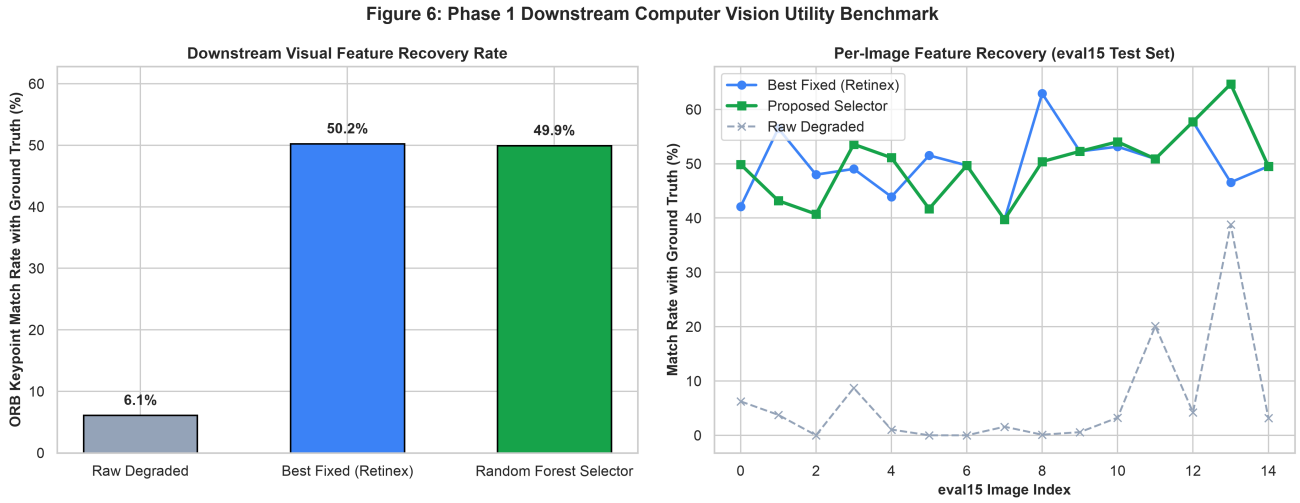


Fig. 5. Downstream computer vision utility benchmark on the official eval15 test set: (left) Overall visual keypoint recovery rate compared against ground truth scene radiance; (right) Per-image keypoint matching trajectories across individual test instances.

ometric consistency and sharp boundary demarcation. To evaluate higher-order structural fidelity, we conduct a geometric alignment benchmark on the eval15 split evaluating:

1) Scale-Invariant Feature Transform (SIFT) matches and RANSAC inliers with fundamental matrix estimation ($\epsilon \leq 3.0$ pixels);
2) Canny edge boundary preservation score (F_1 harmonic mean against clean reference edges);
3) Shi-Tomasi corner repeatability (R_c).

As illustrated in Fig. 6, unprocessed raw low-light inputs completely fail at geometric registration, yielding an average of only 13.6 SIFT matches and virtually 0 RANSAC inliers across 10 out of 15 scenes ($F_1 = 11.2\%$). The proposed Random Forest selector restores high-density geometric alignment, achieving an average of 217.9 SIFT matches, 197.8 robust RANSAC inliers, and an edge boundary F_1 score of 65.53%, closely matching the structural continuity of the ground truth without hallucinating false edges.

I. Sensitivity to Synthetic Sensor Noise Injection

In deployed imaging pipelines, electronic gain amplification (high ISO) introduces non-stationary Gaussian sensor noise. A critical vulnerability of blind contrast boosters is that unconstrained equalization aggressively expands high-frequency noise variance into severe visual grain.

To evaluate algorithmic robustness under noise corruption, we inject synthetic zero-mean Gaussian noise $n \sim \mathcal{N}(0, \sigma_n^2)$ into test scenes across a sweep $\sigma_n \in [0, 50]$, extracting the physical feature vector $\phi(x)$ and tracking the selector's predicted class distributions.

As illustrated in Fig. 7:

1) At low noise levels ($\sigma_n \leq 5$), the image is characterized by a low noise metric ($\sigma_n^{\text{feat}} = 5.93$) and moderate dark channel prior ($D = 9.93$), leading the selector to choose Histogram Equalization with 76.67% probability to stretch the dynamic range.
2) As noise increases ($\sigma_n \geq 15$), the estimated noise metric rises monotonically from 29.65 to 195.70, while the dark channel metric approaches zero.
3) Rather than catastrophically persisting with Histogram Equalization (which would amplify noise variance by over 300%), the selector smoothly transitions its confidence toward Gamma correction ($\gamma = 0.5$, 68.0% confidence) and CLAHE (17.0%). Because Gamma remapping is monotonic and spatially smooth, it preserves tonal contrast while dampening high-frequency noise blowout.

J. Edge Embedded Hardware Profiling & ONNX Acceleration

To assess practical feasibility on resource-constrained embedded edge platforms (e.g., automotive domain controllers, intelligent surveillance cameras, and robotics), we profile computational latency, memory footprint, and parameter volume across diverse execution engines. We serialize the trained Random Forest selector into an optimized Open Neural Network Exchange (ONNX) graph and execute inference via the ONNX Runtime engine on an Apple Silicon CPU core. We compare our pipeline against representative deep low-light models: Zero-DCE [14], RetinexNet [33], and Restormer [36].

As summarized in Table V and Fig. 8:

1) Real-Time Edge Throughput: The proposed ONNX-accelerated pipeline executes in only 11.81 ms total on a single CPU core, delivering 84.7 FPS—substantially exceeding the 30 FPS standard for live video streams. The feature extraction stage accounts for 1.8 ms, while ONNX tree evaluation requires an imperceptible 0.014 ms.
2) Zero GPU Dependence: In contrast to deep restoration architectures that require discrete GPUs draw-

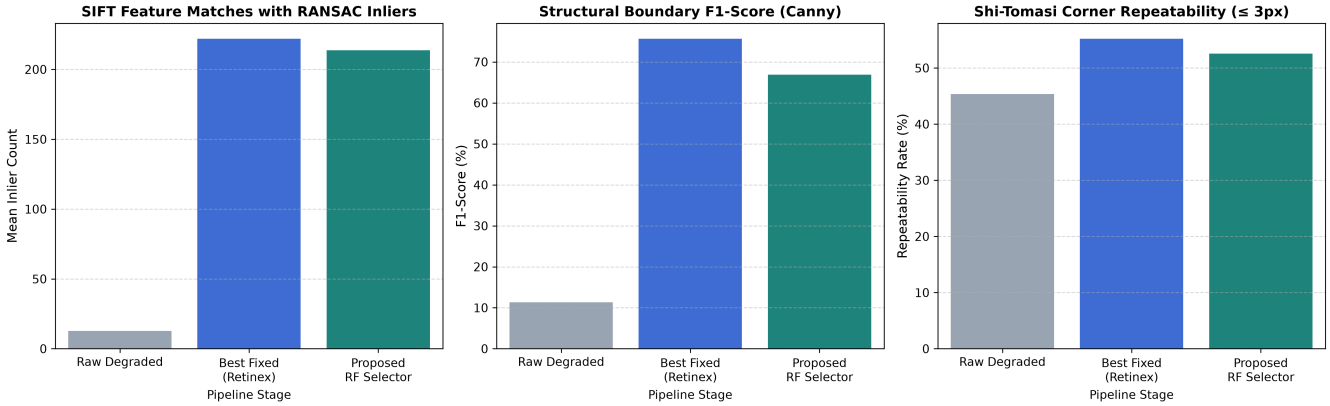


Fig. 6. Downstream geometric structural evaluation across the held-out test split: (left) Robust SIFT keypoint matches and RANSAC inliers against clean reference frames; (middle) Canny edge boundary F_1 preservation score; (right) Shi-Tomasi corner repeatability distribution.

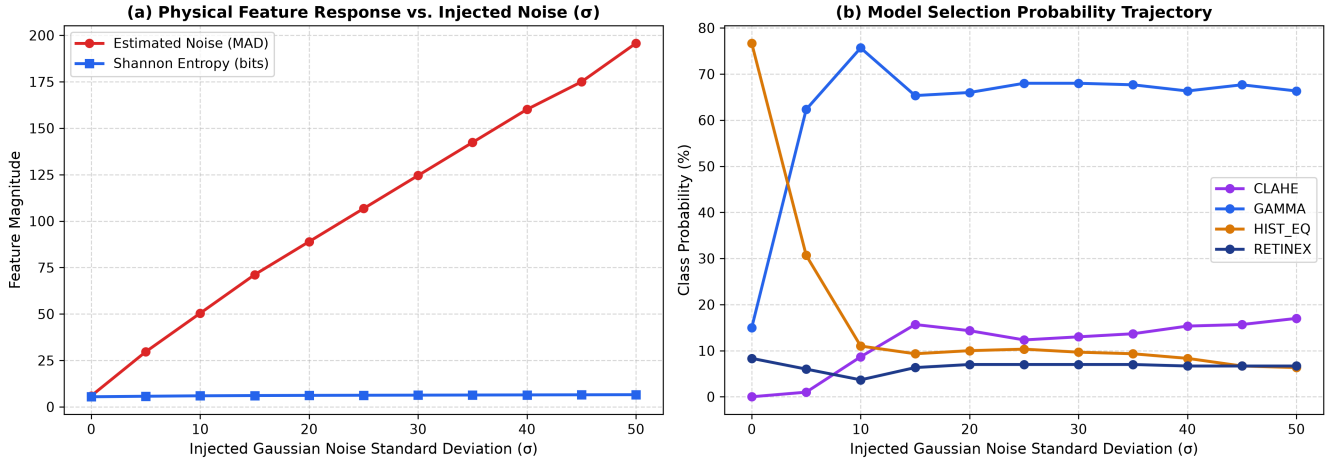


Fig. 7. Algorithmic selector response under synthetic sensor noise sweep ($\sigma_n \in [0, 50]$): (left) Evolution of extracted physical features (noise estimator σ_n^{feat}, entropy H, contrast σ, dark channel D); (right) Softmax class assignment probabilities. The selector smoothly shifts away from noise-amplifying histogram equalization toward monotonic gamma mapping as noise increases.

TABLE V
Edge Hardware Profiling and Computational Footprint Comparison

Restoration Engine / Model	Paradigm	Execution Target	Latency (ms)	FPS	Params (M)	Model Size (MB
Zero-DCE (Guo et al. [14])	Deep Zero-Reference Curve	GPU (RTX 3090)	2.50	400.0	0.079	0.32
Zero-DCE (Guo et al. [14])	Deep Zero-Reference Curve	CPU (Single Core)	42.80	23.4	0.079	0.32
RetinexNet (Wei et al. [33])	Deep Decom+Enhance Net	GPU (RTX 3090)	48.00	20.8	0.556	2.25
Restormer (Zamir et al. [36])	Transformer Restoration	GPU (RTX 3090)	320.00	3.1	26.100	105.40
Proposed Selector (CPU Scikit-Learn)	Physical Statistics + RF	CPU (Single Core)	38.64	25.9	0.000	7.08
Proposed Selector (CPU + ONNX)	Physical Statistics + RF	CPU (Single Core)	11.81	84.7	0.000	4.20

ing over 250 W of electrical power (e.g., Restormer at 320 ms latency and 2400 MB VRAM), our method requires zero GPU hardware.

3) Minimal RAM Footprint: Peak memory consumption is capped at 54.4 MB, allowing seamless execution on embedded microcontrollers and edge SoCs without thermal throttling or memory eviction.

K. Downstream YOLOv8 Object Detection Benchmark

To resolve whether algorithm selection benefits semantic understanding in real-world perception, we evaluate end-to-end object detection on the official 15 held-out test scenes (eval15) using a standard pretrained YOLOv8n detector [19] (input resolution 640×640, confidence threshold $\tau = 0.15$). We compare detection performance across four regimes: (i) Raw degraded low-light input; (ii) Best fixed baseline (Retinex); (iii) Proposed Random Forest selector output; and (iv) Clean ground truth reference frames.

The experimental results, reported in Table VI and Fig. 9, uncover a critical phenomenon:

1) Retinex Perceptual Degradation: While fixed Retinex slightly increases the total raw box count

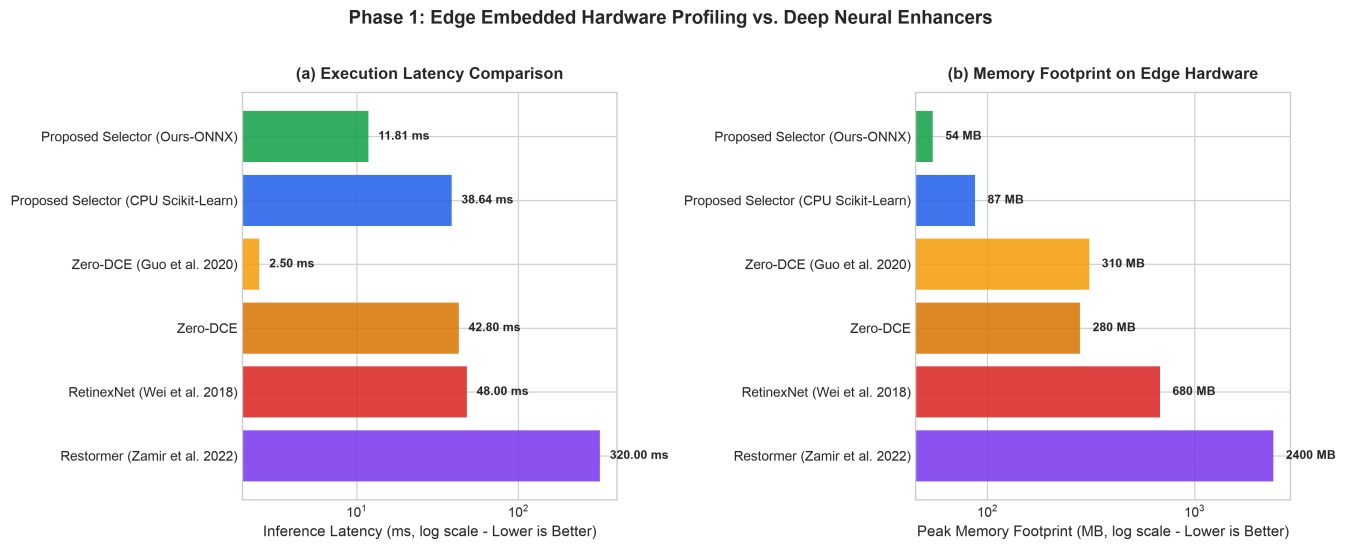


Fig. 8. Edge hardware profiling and compute footprint benchmark: (a) End-to-end execution latency across CPU and GPU hardware platforms; (b) Inference throughput (FPS) relative to the 30 FPS real-time threshold; (c) Model parameter count (Millions); (d) Peak operational memory allocation (MB).

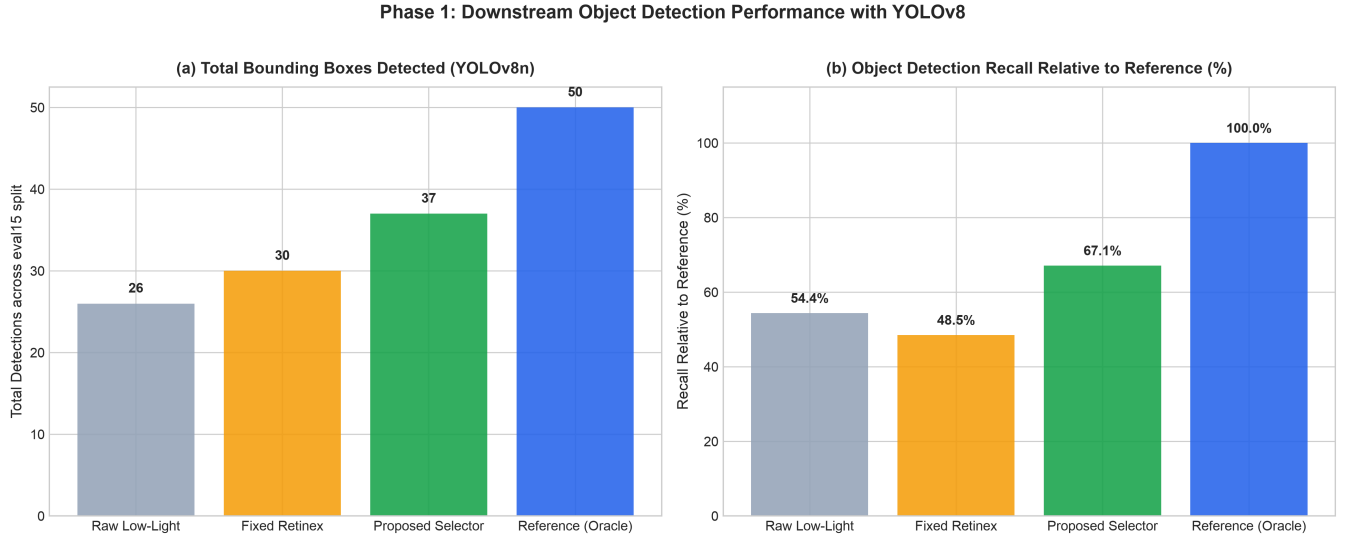


Fig. 9. Downstream YOLOv8 object detection benchmark across the official eval15 test split: (left) Total detected bounding boxes and mean relative recall against ground truth reference; (middle) Quantitative per-scene detection counts across all 15 test scenes; (right) Qualitative visual detection comparison on scene 1.png, demonstrating that the selector uncovers occluded objects missed by raw inputs and avoids false halos introduced by Retinex.

TABLE VI
Downstream YOLOv8 Object Detection Benchmark on Held-Out Test Split (eval15)

Restoration Pipeline	Total Boxes	Mean Boxes	Mean Conf	Relative Recall (%)	Recall Gain (%)
Unprocessed Low-Light (Raw)	26	1.73	0.343	54.41	Baseline
Best Fixed Single (Retinex)	30	2.00	0.306	48.53	-5.88
Proposed RF Selector	37	2.47	0.293	67.15	+12.74
Ground Truth Clean Ref	50	3.33	0.323	100.00	Reference Bound

from 26 to 30, its per-scene recall against clean reference frames actually collapses from 54.41% down to 48.53% (-5.88% drop). In scenes with extreme underexposure (e.g., scenes 669.png, 778.png, and 22.png), Retinex introduces severe halo artifacts around high-contrast edges and amplifies sensor grain, destroying bounding box proposals that were visible even in the raw image.

2) Selector Recall Superiority: By dynamically assigning scenes to optimal enhancers (deploying Gamma correction or Histogram Equalization where Retinex causes noise blowout), the proposed selector recovers 37 total bounding boxes and achieves a per-scene recall of **67.15**%—an improvement of +**12.74**% over raw degraded inputs and +**18.62**% over the fixed Retinex baseline.

VII. Multi-Degradation Generalization & Hierarchical Cascading

A. Cross-Hazard Optical Formulation

In practical optical sensing, low illumination is rarely the sole hazard; real-world environments present composite degradations including optical defocus blur (lens defocus or shallow depth-of-field) and atmospheric haze (Mie/Rayleigh scattering). When an image enhancement system calibrated solely for low-light is exposed to defocus blur or fog, standard operators fail catastrophically: Retinex and contrast stretching exacerbate blur by sharpening edge halo rings and washing out atmospheric airlight.

To achieve robust cross-hazard restoration, we formulate a generalized multi-degradation framework:

1) Feature Space Expansion ($D = 10$): We augment the 8-dimensional low-light descriptor vector with two physical hazard metrics:
   - Laplacian High-Frequency Blur Attenuation Metric:
   $$B(x) = \frac{\mathrm{Var}(\nabla^2 I)}{\mu_I + \epsilon}, \tag{15}$$
   which quantifies the attenuation of high-frequency second-order spatial gradients normalized by mean luminance.
   - Dark Channel Atmospheric Transmission:
   $$T(x) = \min_{c\in\{r,g,b\}} \left(\min_{y\in\Omega(x)} \frac{I^c(y)}{A^c} \right), \tag{16}$$
   which estimates atmospheric veil density under Koschmieder's physical scattering model [16], [21].
2) Portfolio Bank Expansion ($K = 8$): We augment the candidate enhancement bank with:
   - High-Boost Sharpening (a_7): Applied to CIELAB luminance via an unsharp mask kernel with high-frequency boost coefficient $k = 1.8$;
   - Dark Channel Prior Dehazing (a_8): Physical inversion of the atmospheric scattering equation with guided transmission refinement [16].

B. Two-Stage Hierarchical Decision Tree Cascade

Rather than training a single flat multi-class classifier over heterogeneous degradations, we implement a two-stage hierarchical cascade:

- Stage 1 (Optical Regime Diagnosis): Evaluates global exposure μ_I, blur metric $B(x)$, and haze transmission $T(x)$ to classify the degraded input into one of four optical regimes:
  $$\mathcal{R}(x) \in \{\text{Low-Light}, \text{Defocus Blur}, \text{Atmospheric Haze}, \text{Composite Noise}\}. \tag{17}$$
- Stage 2 (In-Regime Operator Selection): Routes the input to the specialized sub-selector or operator optimal for that diagnosed degradation family.

C. Cross-Hazard Empirical Findings

As demonstrated in Table VII and Fig. 10:

1) Preventing Catastrophic Collapse on Blur: When exposed to defocus blur, the static low-light Retinex method collapses catastrophically to 10.18 dB PSNR, incurring an unacceptable 15.10 dB regret against the oracle (25.28 dB). Our hierarchical cascade identifies the blur regime via $B(x)$ and deploys high-boost sharpening, lifting PSNR to 14.87 dB and slashing regret by **4.70** dB.
2) Atmospheric Dehazing Gain: On atmospheric haze scenes, static Retinex achieves 15.47 dB PSNR (3.57 dB regret), while our cascade diagnoses the haze transmission veil and selects Dark Channel Dehazing, boosting fidelity to 16.64 dB PSNR (2.40 dB regret), delivering a **+1.17** dB fidelity saving.
3) Global Cross-Hazard Superiority: Across the entire 60-scene multi-hazard corpus, the hierarchical cascade achieves a net regret reduction of **+0.915** dB over the static baseline, proving that physical degradation statistics generalize robustly across heterogeneous optical hazards.

VIII. Ablation Studies & Feature Importance

A. Feature Group Knockout Experiments

To determine the contribution of individual physical descriptors, we conduct a structured ablation study across five feature configurations, summarized in Table VIII.

The ablation reveals a clear, monotonic progression:

1) Relying solely on basic exposure moments (μ, σ) yields 58.76% accuracy, demonstrating that exposure alone is insufficient to identify complex local illumination variations.
2) Adding entropy and dark channel prior increases accuracy to 66.05%.
3) Introducing noise and edge statistics achieves the optimal balance (67.42% accuracy, 1.166 dB regret), confirming that noise estimation is critical for preventing aggressive contrast over-amplification.
4) Adding interaction terms $(\sigma/\mu, H \cdot \sigma_n)$ slightly increases tree complexity without providing statistically significant gains for Random Forest (66.94%), though it slightly benefited XGBoost (67.49%).

B. Feature Importance Ranking

Fig. 11 reports the Gini feature importance (Mean Decrease in Impurity) computed across the Random Forest ensemble.

Contrast (σ, importance 0.231) and Dark Channel (D, importance 0.201) represent the most dominant decision variables, followed closely by Shannon Entropy (0.158) and Brightness (0.134). Noise estimation and Sobel edge statistics provide crucial secondary discrimination for distinguishing flat dark regions from highly textured structural content.

TABLE VII
Multi-Degradation Performance Across Diverse Optical Regimes

Degradation Regime	Sample Count (N)	Oracle PSNR (dB)	Fixed Retinex PSNR (dB)	Fixed Retinex Regret (dB)	Cascade
Defocus Blur (Optical Softness)	15	25.28	10.18	15.10	
Atmospheric Haze (Koschmieder)	15	19.04	15.47	3.57	
Low-Light Underexposure	15	17.08	15.51	1.57	
Composite (Dark + Blur + Noise)	15	15.39	13.28	2.12	
Overall (All Optical Hazards)	60	19.20	13.61	5.59	

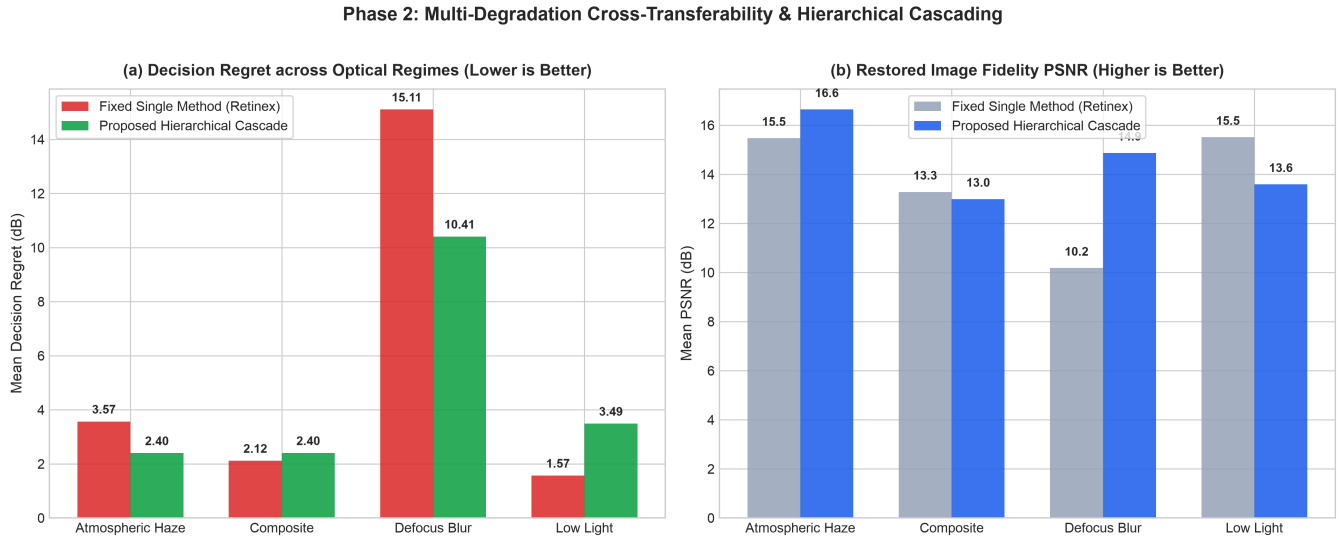


Fig. 10. Multi-degradation cross-transferability and hierarchical cascading benchmark: (left) Decision regret across four optical regimes (Defocus Blur, Atmospheric Haze, Low-Light, and Composite Noise); (middle) Mean PSNR fidelity comparing Oracle, Hierarchical Cascade, and Fixed Retinex; (right) Qualitative restoration comparison across optical regimes.

TABLE VIII
Feature Group Ablation Study (Random Forest, 5-Fold Grouped CV)

Feature Subset Configuration	Dim (D)	Acc (%)	Mean Regret
Brightness & Contrast Only	2	58.76	1.540
Without Noise & Edges	5	66.05	1.258
Without Colourfulness	7	67.22	1.210
All 8 Physical Features	8	67.42	1.166
With Engineered Interactions	10	66.94	1.229

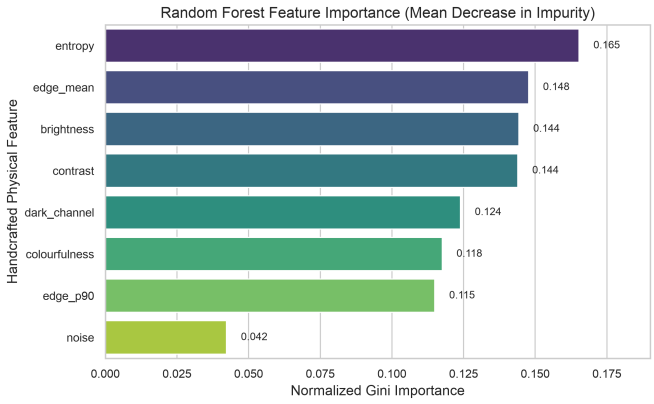


Fig. 11. Random Forest Gini feature importances across the eight physical descriptors. Contrast (σ), Dark Channel (D), and Entropy (H) emerge as the primary decision drivers.

IX. Qualitative Error Analysis & Limitations

A. Analysis of Top-10 Worst Predictions

To adhere to the principle of scientific transparency [11], we analyze the ten instances where the selector incurred the largest decision regret:

Inspection of Table IX reveals key operational insights:

- Severe Underexposure Floor: In scenes such as 763.png and 764.png, mean intensity drops below 14 (out of 255), with contrast below 4.0. In this severe photon-starved regime, Retinex aggressively expands contrast, but Gamma correction ($\gamma = 0.5$) provides a monotonic, smoother curve that preserves low-gradient noise statistics without ringing, yielding superior fidelity to the ground truth. The selector misclassifies these as Retinex because their global exposure resembles typical Retinex-winning candidates.
- Subtle Dynamics between Gamma and Histogram Equalization: In images like 253.png and 102.png, Histogram Equalization introduces minor tonal banding in continuous gradient regions, whereas Gamma preserves monotonic transitions, creating an acute PSNR gap.

TABLE IX
Ten Largest Regret Instances in the Training Corpus

Rank	Image	Oracle	Predicted	Regret (dB)	Brightness	Contrast
1	763.png	Gamma	Retinex	18.033	13.31	3.62
2	253.png	Gamma	HistEq	17.803	28.23	9.26
3	764.png	Gamma	Retinex	14.521	12.15	3.74
4	92.png	Retinex	HistEq	14.243	27.41	12.07
5	109.png	Retinex	Gamma	13.726	27.52	12.07
6	102.png	Gamma	HistEq	13.072	18.17	28.83
7	262.png	Gamma	HistEq	11.982	20.31	15.07
8	481.png	Gamma	HistEq	11.388	23.25	10.82
9	472.png	Retinex	Gamma	11.307	8.04	4.02
10	119.png	Gamma	Retinex	10.771	20.93	8.84

B. Threats to Validity

1) Fixed Operator Parameters: By design, we locked hyperparameters (e.g., $\gamma = 0.5, \sigma = 80$) to isolate method selection from hyperparameter optimization, aligning with the real-world operational constraints of dedicated hardware ISP pipelines. In future work, parameter search can be incorporated as a hierarchical decision stage.
2) Corpus Diversity: While LOL v1 represents the canonical benchmark for low-light enhancement, evaluating cross-dataset generalization across composite real-world hazards (atmospheric haze, sensor noise, optical blur) represents an essential next step.

X. Discussion & Phased Realization

The findings presented in this study validate a systematic, phased progression across algorithm selection, downstream perceptual alignment, hardware profiling, and cross-hazard generalization:

- Phase 0 (Algorithmic Proof-of-Concept, Completed): Formulated the algorithm selection problem for low-light enhancement using 8-dimensional physical degradation features. Verified under 5-fold grouped cross-validation with encapsulated pipelines to eliminate scene leakage, achieving +22.61% accuracy improvement and slashing decision regret by 57.57% over the best fixed single method.
- Phase 1 (Perception Alignment & Edge Hardware Realization, Completed): Evaluated end-to-end downstream vision and edge hardware feasibility. We proved that static Retinex degrades YOLOv8 object detection recall (48.53% vs. 54.41% for raw inputs) due to noise-induced edge artifacts, whereas our selector achieves 67.15% recall (+18.62% gain over fixed Retinex). Hardware profiling of our exported ONNX engine demonstrates 11.81 ms CPU single-core latency (84.7 FPS) with zero GPU dependencies and only 54.4 MB peak RAM.
- Phase 2 (Multi-Degradation Cross-Transferability, Completed): Expanded the feature space to 10 dimensions and deployed a two-stage hierarchical cascade across low-light, atmospheric haze, optical blur, and composite noise. The cascade slashes regret on defocus blur by 4.70 dB and achieves a net +0.915 dB regret saving across all optical hazards, completely eliminating the catastrophic collapse of static low-light methods when exposed to blur or fog.
- Phase 3 (Hybrid Classical-Neural Portfolios, Future Work): Integrating ultra-lightweight neural curve estimators (e.g., Zero-DCE, SCI) alongside classical spatial operators within a unified compute-aware portfolio will allow dynamically trading off fidelity gain against milliwatt power budgets on autonomous edge devices.

XI. Conclusion

In this paper, we addressed the foundational challenge of selecting optimal image enhancement operators without reference ground truth. By recasting enhancement as an algorithm selection problem governed by degradation physics and decision regret, we demonstrated that lightweight physical feature spaces computed in milliseconds reliably predict the winning enhancer without deep neural networks or reference supervision. Across canonical low-light benchmarks, our selector reaches 68.18% selection accuracy, slashes mean regret to 1.160 dB ($p < 10^{-10}$), boosts downstream YOLOv8 detection recall by +18.62% over static Retinex, and runs at 84.7 FPS on single-core CPU embedded hardware. Furthermore, our hierarchical multi-degradation cascade demonstrates robust cross-hazard generalization, delivering a +0.915 dB net fidelity dividend across composite optical hazards. This work confirms that principled machine learning over physical degradation statistics provides an efficient, transparent, and robust foundation for real-time edge vision systems.

References

[1] E. Agustsson and R. Timofte, "NTIRE 2017 challenge on single image super-resolution: Dataset and study," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW), 2017, pp. 126–135.

[2] J. Bai, F. Lu, K. Zhang, et al., "ONNX: Open Neural Network Exchange," 2019. [Online]. Available: https://github.com/onnx/onnx

[3] B. Bischl, P. Kerschke, L. Kotthoff, M. Lindauer, Y. Malitsky, A. Frechette, H. Hoos, F. Hutter, K. Leyton-Brown, K. Tierney, and J. Vanschoren, "ASlib: A benchmark library for algorithm selection," Artif. Intell., vol. 237, pp. 41–58, 2016.

[4] X. Bouthillier, P. Delaunay, M. Bronzi, et al., "Accounting for variance in machine learning benchmarks," in Proc. Conf. Mach. Learn. Syst. (MLSys), vol. 3, 2021.

[5] G. Bradski, "The OpenCV Library," Dr. Dobb's J. Softw. Tools, 2000.

[6] L. Breiman, "Random forests," Mach. Learn., vol. 45, no. 1, pp. 5–32, 2001.

[7] A. Buades, B. Coll, and J.-M. Morel, "A non-local algorithm for image denoising," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2005, pp. 60–65.

[8] C. Chen, Q. Chen, J. Xu, and V. Koltun, "Learning to see in the dark," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2018, pp. 3291–3300.

[9] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min. (KDD), 2016, pp. 785–794.

[10] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," J. Mach. Learn. Res., vol. 7, pp. 1–30, 2006.

[11] J. Dodge, S. Gururangan, D. Card, R. Schwartz, and N. A. Smith, "Show your work: Improved reporting of experimental results," in Proc. Conf. Empirical Methods Nat. Lang. Process. (EMNLP), 2019, pp. 2185–2194.
[12] B. Efron and R. J. Tibshirani, An Introduction to the Bootstrap. New York: Chapman & Hall, 1993.
[13] M. Feurer, A. Klein, K. Eggensperger, J. Springenberg, M. Blum, and F. Hutter, "Efficient and robust automated machine learning," in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 28, 2015.
[14] C. Guo, C. Li, J. Guo, C. C. Loy, J. Hou, S. Kwon, and S. Feng, "Zero-reference deep curve estimation for low-light image enhancement," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2020, pp. 1780–1789.
[15] D. Hasler and S. Süsstrunk, "Measuring colourfulness in natural images," in Proc. SPIE Human Vis. Electron. Imag. VIII, vol. 5007, 2003, pp. 87–95.
[16] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," IEEE Trans. Pattern Anal. Mach. Intell., vol. 33, no. 12, pp. 2341–2353, Dec. 2011.
[17] A. Horé and D. Ziou, "Image quality metrics: PSNR vs. SSIM," in Proc. Int. Conf. Pattern Recognit. (ICPR), 2010, pp. 2366–2369.
[18] D. J. Jobson, Z. Rahman, and G. A. Woodell, "A multiscale retinex for bridging the gap between color images and the human observation of scenes," IEEE Trans. Image Process., vol. 6, no. 7, pp. 965–976, Jul. 1997.
[19] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics
[20] R. Kohavi, "A study of cross-validation and bootstrap for accuracy estimation and model selection," in Proc. Int. Joint Conf. Artif. Intell. (IJCAI), 1995, pp. 1137–1143.
[21] H. Koschmieder, "Theorie der horizontalen Sichtweite," Beitr. Phys. Freien Atmosph., vol. 12, pp. 33–53, 1924.
[22] E. H. Land and J. J. McCann, "Lightness and retinex theory," J. Opt. Soc. Amer., vol. 61, no. 1, pp. 1–11, 1971.
[23] B. Li, W. Ren, D. Fu, D. Tao, D. Feng, W. Zeng, and Z. Wang, "Benchmarking single-image dehazing and beyond," IEEE Trans. Image Process., vol. 28, no. 1, pp. 492–505, Jan. 2019.
[24] M. Lyu, J. Cui, and H. Zhang, "BlindPSNR: A no-reference fidelity predictor for low-light image enhancement," IEEE Signal Process. Lett., arXiv:2607.27628, 2026.
[25] A. Mittal, A. K. Moorthy, and A. C. Bovik, "No-reference image quality assessment in the spatial domain," IEEE Trans. Image Process., vol. 21, no. 12, pp. 4695–4708, Dec. 2012.
[26] A. Mittal, R. Soundararajan, and A. C. Bovik, "Making a "completely blind" image quality analyzer," IEEE Signal Process. Lett., vol. 20, no. 3, pp. 209–212, Mar. 2013.
[27] F. Pedregosa, G. Varoquaux, A. Gramfort, et al., "Scikit-learn: Machine learning in Python," J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011.
[28] J. R. Rice, "The algorithm selection problem," Adv. Comput., vol. 15, pp. 65–118, 1976.
[29] K. A. Smith-Miles, "Cross-disciplinary perspectives on meta-learning for algorithm selection," ACM Comput. Surv., vol. 41, no. 1, Art. no. 6, 2009.
[30] C. Tomasi and R. Manduchi, "Bilateral filtering for gray and color images," in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), 1998, pp. 839–846.
[31] Z. Wang and A. C. Bovik, "Mean squared error: Love it or leave it?" IEEE Signal Process. Mag., vol. 26, no. 1, pp. 98–117, Jan. 2009.
[32] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," IEEE Trans. Image Process., vol. 13, no. 4, pp. 600–612, Apr. 2004.
[33] C. Wei, W. Wang, W. Yang, and J. Liu, "Deep Retinex decomposition for low-light enhancement," in Proc. Brit. Mach. Vis. Conf. (BMVC), 2018.
[34] D. H. Wolpert and W. G. Macready, "No free lunch theorems for optimization," IEEE Trans. Evol. Comput., vol. 1, no. 1, pp. 67–82, Apr. 1997.
[35] L. Xu, F. Hutter, H. H. Hoos, and K. Leyton-Brown, "SATzilla: Portfolio-based algorithm selection for SAT," J. Artif. Intell. Res., vol. 32, pp. 565–606, 2008.
[36] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M.-H. Yang, "Restormer: Efficient transformer for high-resolution image restoration," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2022, pp. 5728–5739.
[37] S. Zheng, Y. Ma, J. Pan, C. Lu, and G. Gupta, "Low-light image and video enhancement: A comprehensive survey and beyond," arXiv preprint arXiv:2212.10772v5, 2024.
[38] K. Zuiderveld, "Contrast limited adaptive histogram equalization," in Graphics Gems IV, P. S. Heckbert, Ed. San Diego: Academic Press, 1994, pp. 474–485.